Resonance

Pupil sheet

Name: _____ **Date:** _____

Safety: I hold the box with both hands, with a metre of space around me; I never throw it or let it drop.

1. What I see on the screen

Three buildings, each with **its own rhythm** written just below it: the tower, $T = 1.0$ s; the block, $T = 0.5$ s; the house, $T = 0.3$ s.

The box is held flat and moved from side to side, never tilted. **The star rule:** it lights up only when *one* building dances and the other two stay still. **Shaking harder does not win it** — why not is today's question.

2. What I do on my own

First I predict. I do not change my prediction after I play.
Shaking harder will make: ☐ more buildings dance ☐ fewer ☐ the same number
The slowest rhythm belongs to: ☐ the tower ☐ the block ☐ the house

because _____

_____

Which rhythm wakes which building. Box flat, moved sideways, same rhythm for **at least five seconds** — the dancing needs time to grow.

How I move the box	Building that grows	Star?
Slowly, 1 s there and back		yes / no
Twice as fast		yes / no
Three times as fast		yes / no

The test: I shake as hard as I can, for thirty seconds, with no rhythm. **No stars is a result.**

What I do	Bars that rise	Stars I win
Shake hard, no rhythm		
Move little, but in rhythm		

The hard question. What mattered more: ☐ how hard ☐ how steadily I shook

3. What I do on my own

From a period to beats per minute. A period is a **complete there-and-back**: left, right, back again — **not just one way**. That is where most people go wrong, ending up twice too fast. The frequency is $f = 1/T$, and the beats per minute are $60 \times f$.

Building	T (s)	f (Hz)	Beats per minute
Tower, 10 storeys	1.0	1.0	60 (worked out)
Block, 5 storeys	0.5		
House, 3 storeys	0.3		

The engineers' rule. A building with N storeys has a period of about $N/10$ seconds. Does it fit the three above? ☐ yes ☐ no

My block has _____ storeys, so its period is about _____ s. The most dangerous rhythm would be _____ beats per minute.

4. The conclusion I write

I predicted that _____

_____

The data show that _____

_____

Tall buildings sway more slowly because _____

_____

Exit ticket

Why does shaking harder not help?

_____